

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 22: Trigonometrical Ratios

Every formula and result of the chapter on one sheet

1	Sine	$\sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
2	Cosine	$\cos \theta = \frac{\text{base}}{\text{hypotenuse}}$
3	Tangent	$\tan \theta = \frac{\text{perpendicular}}{\text{base}}$
4	Cosecant	$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$
5	Secant	$\sec \theta = \frac{1}{\cos \theta}$
6	Cotangent	$\cot \theta = \frac{1}{\tan \theta}$
7	Quotient relation	$\tan \theta = \frac{\sin \theta}{\cos \theta}$
8	Quotient relation	$\cot \theta = \frac{\cos \theta}{\sin \theta}$
9	Square relation	$\sin^2 \theta + \cos^2 \theta = 1$
10	Square relation	$1 + \tan^2 \theta = \sec^2 \theta$
11	Square relation	$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
12	Sides from the hypotenuse	Perpendicular = $h \sin \theta$, base = $h \cos \theta$